

Report: Project 3

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Course: CSCI 441/541

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1. Data Preparation

The German Credit Risk dataset in ARFF format was loaded and processed. The following steps were undertaken:

1. Converted byte strings to standard strings.
2. Applied one-hot encoding for categorical features and normalized numerical features using StandardScaler.
3. Split the dataset into training (80%) and testing (20%) subsets.

Dataset Summary:

- Features: 48 (after encoding)
- Training Samples: 800
- Testing Samples: 200

2. Model Building

Architecture

The MLP model was structured with:

- Input Layer: 48 features
- Hidden Layer 1: 64 neurons, ReLU activation
- Dropout Layer: 30%
- Hidden Layer 2: 32 neurons, ReLU activation
- Dropout Layer: 30%
- Output Layer: 1 neuron, Sigmoid activation (binary classification)

Model Summary:

Layer Name	Output Shape	Parameters
Dense_1	(64,)	3,136
Dropout_1	(64,)	0
Dense_2	(32,)	2,080
Dropout_2	(32,)	0
Output	(1,)	33
Total Params		5,249

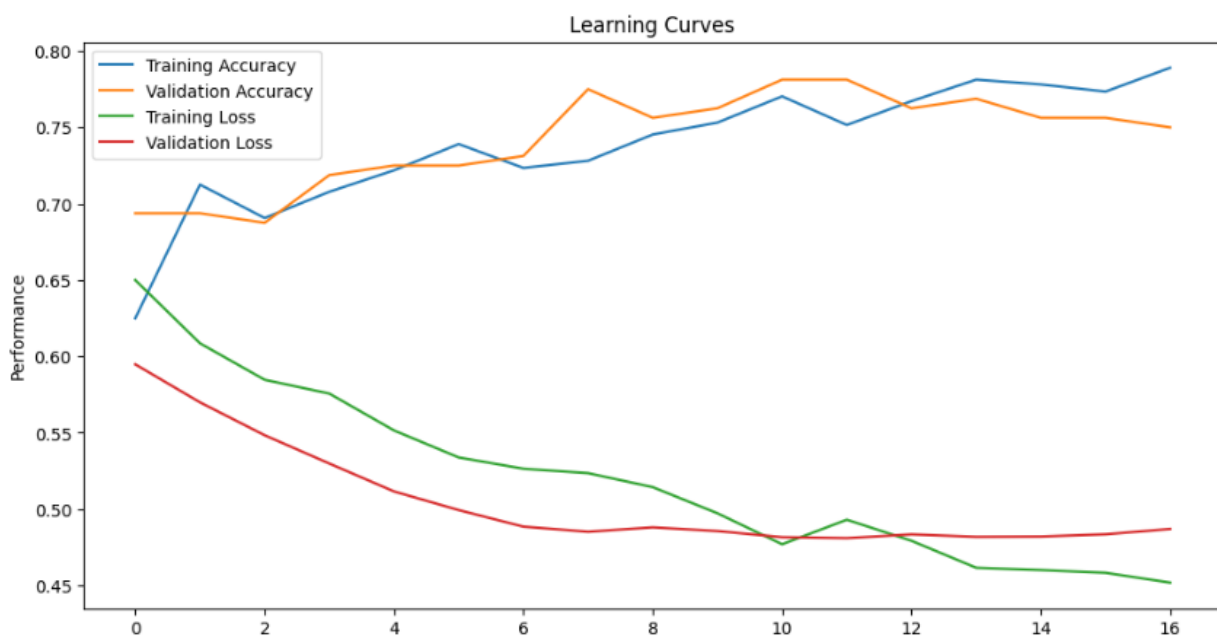
3. Model Training

Training employed:

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Batch Size: 32
- Epochs: 50
- Callbacks: Early stopping and checkpoint saving.

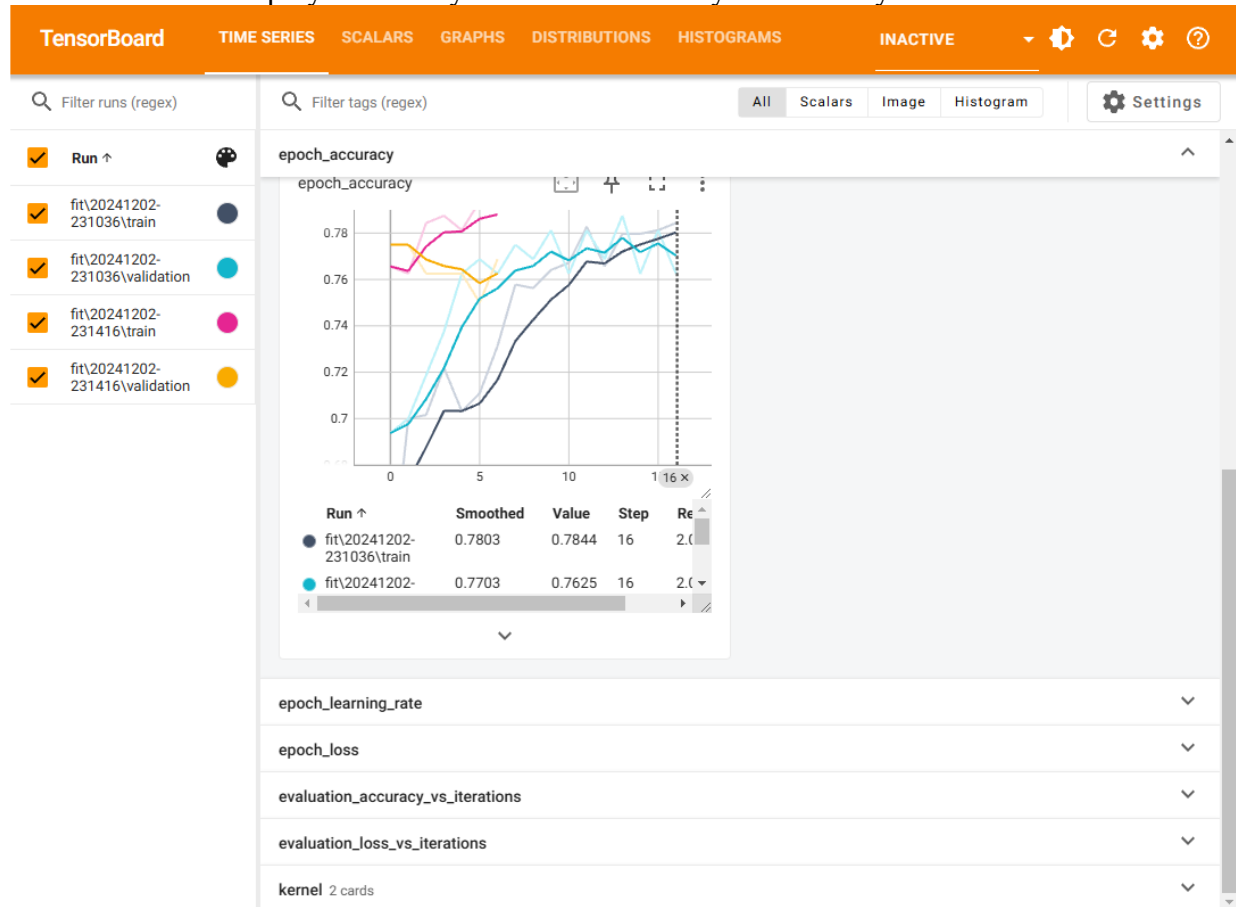
4. Model Evaluation

a) Learning Curves: Learning curves demonstrated consistent improvement in both accuracy and loss, with no significant overfitting



b) TensorBoard Visualization

TensorBoard was employed to analyze loss and accuracy interactively.



c) Analysis

- Training and validation metrics were closely aligned.
- Overfitting was effectively managed through dropout layers and early stopping.

5. Prediction

On the test dataset:

- **Accuracy:** 0.78
- **Loss:** 0.48

Sample Predictions:

Prediction	Actual
1	1
0	0

6. Hyperparameter Tuning

Hyperparameters were optimized using Keras Tuner. Best configuration included:

- Hidden Layer 1 Neurons: 64
- Hidden Layer 2 Neurons: 32
- Dropout Rates: 0.3 for both layers
- Learning Rate: 0.001, unless specified during tuning

Validation accuracy of the tuned model: **0.79**.

7. Conclusion

The project successfully developed a MLP model for credit risk prediction, achieving a test accuracy of 78%, which underscores the model's reliability. The inclusion of dropout layers and early stopping effectively mitigated overfitting, ensuring stable performance. Additionally, the use of Keras Tuner optimized the model, leading to improved accuracy. Future additions could include leveraging ensemble models or transformer-based architectures, as well as incorporating other features to further enhance the models predictive capabilities.